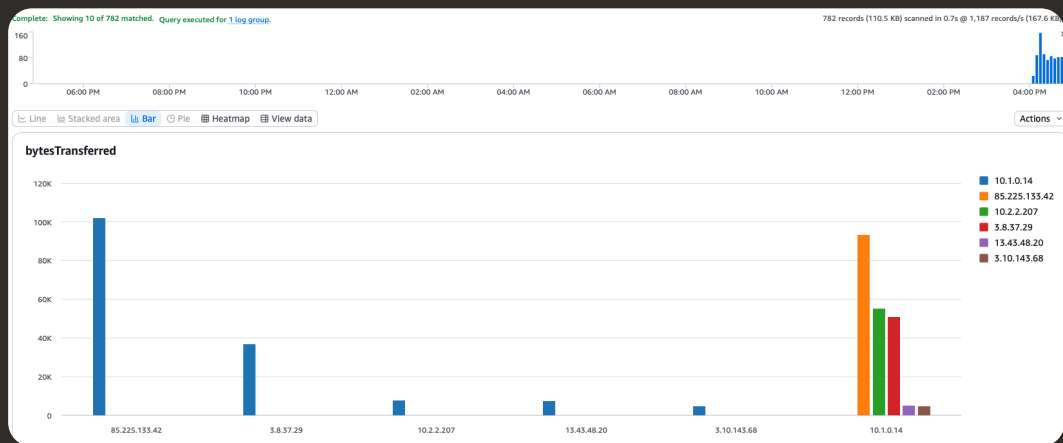




VPC MONITORING WITH FLOW LOGS

By Nabil Ahmed | September 2026

VPC Monitoring with Flow Logs



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a foundational AWS service that lets us control the underlying network for our resources, so that we can control the traffic flow, monitor for security, organise our resources.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create two isolated networks and make them exchange data between them privately and securely with a peer connection and monitoring the data being sent between them using Flow Logs in CloudWatch.

One thing I didn't expect in this project was...

One thing I didn't expect in this project was seeing the amount of visualisation possibilities there are with log analytics which makes understanding and interpreting logs a lot more easier.

This project took me...

This project took me 1 hour and 45 minutes including writing up the documentation.

In the first part of my project...

Step 1 - Set up VPCs

In this step, I will set up two VPCs from scratch. Network monitoring can still be done with just a single VPC, but I think its better to have the extra challenge and tackle VPC peering in this project too.

Step 2 - Launch EC2 instances

In this step, I will launch two EC2 instance - one in each VPC. Doing this is important to set up the remainder of my project as my EC2 instances will generate traffic that VPC Flow Logs will monitor.

Step 3 - Set up Logs

In this step, I will be setting up VPC Flow Logs to start monitoring network traffic. I am also setting up a storage space for my flow logs.

Step 4 - Set IAM permissions for Logs

In this step, I will provide VPC Flow Logs with the permissions to create logs and upload them into my log group in CloudWatch.

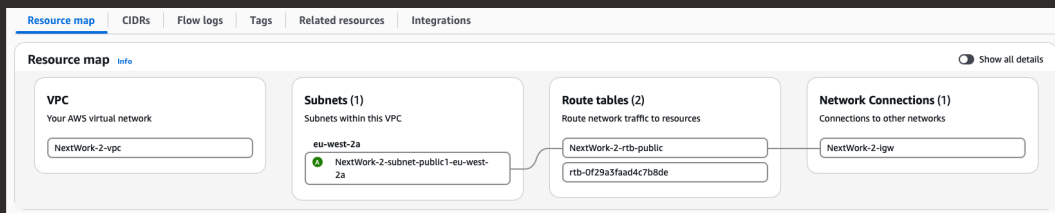
Multi-VPC Architecture

I started my project by launching two VPCs. I created two public subnets (one public subnet in each VPC) with no private subnets.

The CIDR blocks for VPCs 1 and 2 are 10.1.0.0/16 and 10.2.0.0/16. They have to be unique because having overlapping CIDR blocks will cause network routing/traffic issues down the line when traffic is needing to go from one VPC to another.

I also launched EC2 instances in each subnet

My EC2 instances' security groups allow SSH and ICMP type traffic. This is because EC2 Instance Connect will need to access my EC2 instance using SSH-type traffic, and because I need to allow ICMP type traffic for connectivity tests later.



Logs

Logs are like diary entries for computer systems. They are detailed records of any kind of activity related to the traffic/resources/AWS service that the log is tracking.

Log groups are groupings of logs i.e. logs that belong to the project/application/source are often in a log group together. Its a way of organising your logs.

Successfully created flow log for the following resource:
vpc-03a06373885d627f3

fl-0cf9d8c30e348fb6d / NextWorkVPCFlowLog

Actions

Details

Flow log ID
fl-0cf9d8c30e348fb6d

Name
NextWorkVPCFlowLog

State
Active

Creation Time
Monday, 7 September 2026 at 17:07:16 BST

Destination Type
cloud-watch-logs

Destination Name
[NextWorkVPCFlowLogGroup](#)

IAM Role
[arn:aws:iam::840032438341:role/NextWorkVPCFlowLogRole](#)

Cross Account IAM Role
-

Traffic Type
All

Max Aggregation Interval
1 minute

Log Format
Default

Tag field specifications
-

File Format
-

Hive Compatible Partitions
-

Partition Logs
-

IAM Policy and Roles

I created an IAM policy so that I can define a rule that allows policy holders e.g. my VPC Flow Logs service the ability to create log streams and upload them into CloudWatch.

I also created an IAM role because services like VPC Flow Logs have to be associated with a role instead of JSON. Creating an IAM role will be necessary to give my VPC Flow Logs the access it needs to record and upload logs.

A custom trust policy is a specific type of policy used for designing who/what is allowed to access the IAM role.

Custom trust policy

Create a custom trust policy to enable others to perform actions in this account.

```
1 {
2   "Version": "2012-10-17",
3   "Statement": [
4     {
5       "Sid": "Statement1",
6       "Effect": "Allow",
7       "Principal": {
8         "Service": "vpc-flow-logs.amazonaws.com"
9       },
10      "Action": "sts:AssumeRole"
11    }
12  ]
13 }
```

In the second part of my project...

Step 5 - Ping testing and troubleshooting

In this step, I will be generating network traffic. This becomes important when I am communicating about cloud networks/ cloud engineering.

Step 6 - Set up a peering connection

In this step, I will set up a peering connection so that VPCs 1 and 2 can talk directly with each other. I will do this by also modifying the route tables to include a route that goes from the VPC to the peer connection.

Step 7 - Analyze flow logs

In this step, I will be tracking the network data that has been collected on my VPCs and then analyse that data to extract insights.

Connectivity troubleshooting

My first ping test between my EC2 instances had no replies, which means ICMP traffic could be blocked by security group/NACLs, or maybe the traffic is being routed to a wrong path.



```
#_
#####
~\#####\
~~\#####|
~~\###|
~~\#/
~~v~'~>
~~~~
~~~.~
~~~/_/
~~~/_m/'~>

Amazon Linux 2023

https://aws.amazon.com/linux/amazon-linux-2023

[ec2-user@ip-10-1-0-14 ~]$ ping 10.2.2.207
PING 10.2.2.207 (10.2.2.207) 56(84) bytes of data.
```

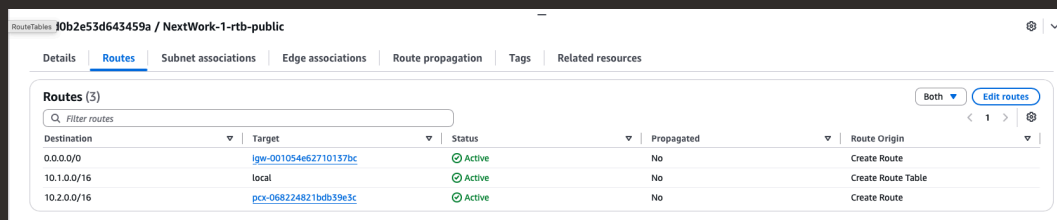
I could receive ping replies if I ran the ping test using the other instance's public IP address, which means the second instance is actually allowing ICMP traffic.

Connectivity troubleshooting

Looking at VPC 1's route table, I identified that the ping test with Instance 2's private address failed because I do not have a route in my VPC's route table that directs traffic from one VPC to another.

To solve this, I set up a peering connection between my VPCs

I also updated both VPCs' route tables so that that traffic from one of the VPCs and heading to the other VPC's private IPV4 address can get directed to go through the peer connection instead of the public internet.



The screenshot shows the AWS Route Tables console for the VPC `10b2e53d643459a` (Name: `NextWork-1-rtb-public`). The `Routes` tab is selected, displaying a table with 3 routes. The table has columns for Destination, Target, Status, Propagated, and Route Origin. The routes are:

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	lgw-001054e62710137bc	Active	No	Create Route
10.1.0.0/16	local	Active	No	Create Route Table
10.2.0.0/16	pcx-068224821bdb39e3c	Active	No	Create Route

Connectivity troubleshooting

I received ping replies from Instance 2's private IP address! This means setting up the peering connection and then the route table solved the connectivity error of the VPC's traffic not being able to navigate from one VPC to another.

```
64 bytes from 10.2.2.207: icmp_seq=545 ttl=127 time=0.190 ms
64 bytes from 10.2.2.207: icmp_seq=546 ttl=127 time=0.187 ms
64 bytes from 10.2.2.207: icmp_seq=547 ttl=127 time=0.162 ms
64 bytes from 10.2.2.207: icmp_seq=548 ttl=127 time=0.217 ms
64 bytes from 10.2.2.207: icmp_seq=549 ttl=127 time=0.169 ms
64 bytes from 10.2.2.207: icmp_seq=550 ttl=127 time=0.166 ms
64 bytes from 10.2.2.207: icmp_seq=551 ttl=127 time=0.168 ms
64 bytes from 10.2.2.207: icmp_seq=552 ttl=127 time=0.169 ms
64 bytes from 10.2.2.207: icmp_seq=553 ttl=127 time=0.169 ms
64 bytes from 10.2.2.207: icmp_seq=554 ttl=127 time=0.196 ms
64 bytes from 10.2.2.207: icmp_seq=555 ttl=127 time=0.162 ms
64 bytes from 10.2.2.207: icmp_seq=556 ttl=127 time=0.185 ms
64 bytes from 10.2.2.207: icmp_seq=557 ttl=127 time=0.204 ms
64 bytes from 10.2.2.207: icmp_seq=558 ttl=127 time=0.205 ms
64 bytes from 10.2.2.207: icmp_seq=559 ttl=127 time=0.235 ms
64 bytes from 10.2.2.207: icmp_seq=560 ttl=127 time=0.213 ms
```

Analyzing flow logs

Flow logs tell us about the source and destination of the network traffic, the amount of data being transferred, whether the traffic was accepted or rejected.

For example, the flow log I've captured tells us that traffic went from 66.132.195.17 to 10.1.0.14 and was rejected. However the recording before that accepted traffic from 10.1.0.14 to 10.2.2.207 and this was accepted by the security groups and NACLs of my VPC.

Log events	
You can use the filter bar below to search for and match terms, phrases, or values in your log events. Learn more about filter patterns	
Filter events - press enter to search Clear1m30m1h	
▶	Timestamp Message
	CloudWatch console feature There are older events to load. Load more.
▼	2026-09-07T16:25:22.000Z2 840032438341 eni-06e8d49e8a1f0bc13 10.1.0.14 10.2.2.207 0 0 1 63 5292 1788798322 1788798348 ACCEPT OK 2 840032438341 eni-06e8d49e8a1f0bc13 10.1.0.14 10.2.2.207 0 0 1 63 5292 1788798322 1788798348 ACCEPT OK
▼	2026-09-07T16:25:22.000Z2 840032438341 eni-06e8d49e8a1f0bc13 66.132.195.17 10.1.0.14 55547 31002 6 1 60 1788798322 1788798348 REJECT OK 2 840032438341 eni-06e8d49e8a1f0bc13 66.132.195.17 10.1.0.14 55547 31002 6 1 60 1788798322 1788798348 REJECT OK

Logs Insights

Logs Insights/Analytics is a special tool within Amazon CloudWatch that helps us with analysing logs and creating visual graphs and charts through queries.

I ran the query 'Top 10 byte transfers by source and destination IP addresses'. This query analyses the flow logs collected on EC2 instance 1, and return the top 10 pairs of IP addresses based on the amount of data transferred between them.

